

● Data-forward

● Policy-relevant

● 1796-2020

The Power of Vaccination: From Smallpox Eradication to COVID- 19 in Record Time

A data-driven overview of how vaccines have transformed public health by eliminating or drastically reducing deadly diseases over the past two centuries.

Audience: public health professionals and policy analysts

Source: Our World in Data (ourworldindata.org/vaccination)



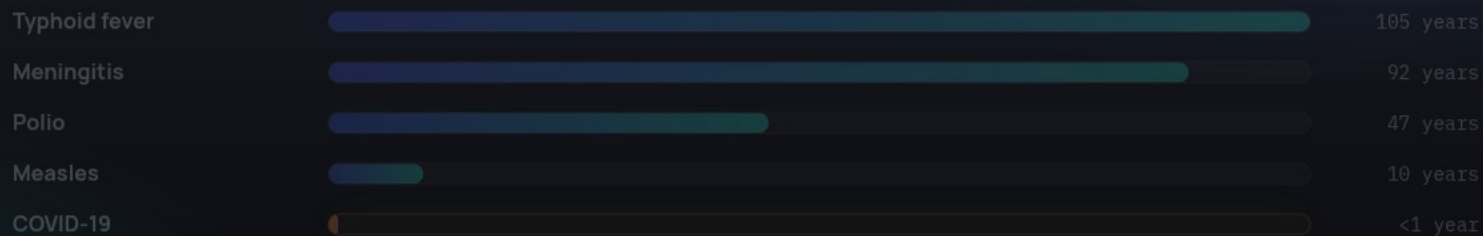
INNOVATION PACE

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

Years from pathogen identification to vaccine licensure (selected examples, 1884–2020).

● Typical

● COVID-19 emphasized



EVIDENCE OF ELIMINATION & REDUCTION

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

3 Diseases with 100% case reduction shown here

Per-million annual cases and deaths before vs. after vaccine introduction (Roush & Murphy, 2007 as cited by Our World in Data).

Disease	Pre-vaccine cases (per million/yr)	Post-vaccine cases (per million/yr)	Case reduction	Pre-vaccine deaths (per million/yr)	Death reduction
Diphtheria	158	0	100% (eliminated)	13.7	100%
Acute polio	141	0	100% (eliminated)	10	100%
Smallpox	250	0	100% (eliminated)	2.9	100%
Pneumococcal	233	139	40.5%	24	31.3%
Acute hepatitis B	273	44	83.9%	1	83.6%

50-STATE SHIFT

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

Heatmap of measles cases per 100,000 by state (1929–2022), with milestone annotations.

1963 →
1980

Milestones referenced
in the heatmap



1963

1971

1980

GLOBAL CAMPAIGN

Smallpox: The Only Human Disease Eradicated Through Vaccination

Key milestones from WHO action to last natural case and certification.



COMMUNITY PROTECTION

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

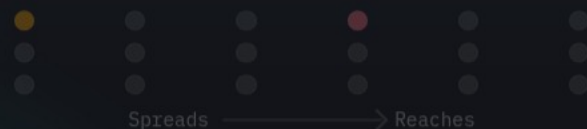
When enough people are vaccinated, pathogens cannot spread effectively — shielding people who cannot be vaccinated.

Vaccinated

Vulnerable

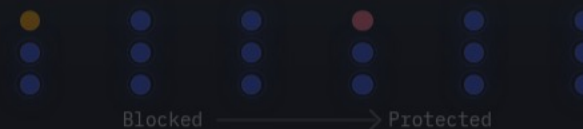
Pathogen

Low coverage: transmission reaches vulnerable



Vulnerable includes newborns, immunocompromised individuals, and cancer patients on chemotherapy.

High coverage: vaccinated “shields” break spread



Population-level protection reduces effective spread, protecting those who cannot be vaccinated.

INCOMPLETE IMPACT

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

Even with vaccines, some diseases retain substantial burden — and global access remains uneven.

40.5%
Pneumococcal case reduction (US, per million/yr)

31.3%
Pneumococcal death reduction

89%
Hepatitis A case reduction

83.9%
Acute hepatitis B case reduction



WHAT TO TAKE FORWARD

Key Takeaways: Vaccines Have Eliminated Deadly Diseases – But the Job Isn't Done

- **100% elimination (US):** Diphtheria, smallpox, and acute polio reach 0 post-vaccine cases with 100% death reduction.
- **Acceleration:** timelines collapse from 105 years (typhoid) to <1 year (COVID-19).
- **Community effect:** herd immunity protects newborns, immunocompromised patients, and people on chemotherapy.
- **Measles:** near-zero across all states after 1980 kindergarten mandate, with sporadic outbreaks.
- **Remaining burden:** pneumococcal (40.5% case reduction), hepatitis A (89%), acute hepatitis B (83.9%).

● Evidence

● Mechanism

● Action

